

July 17, 1934.

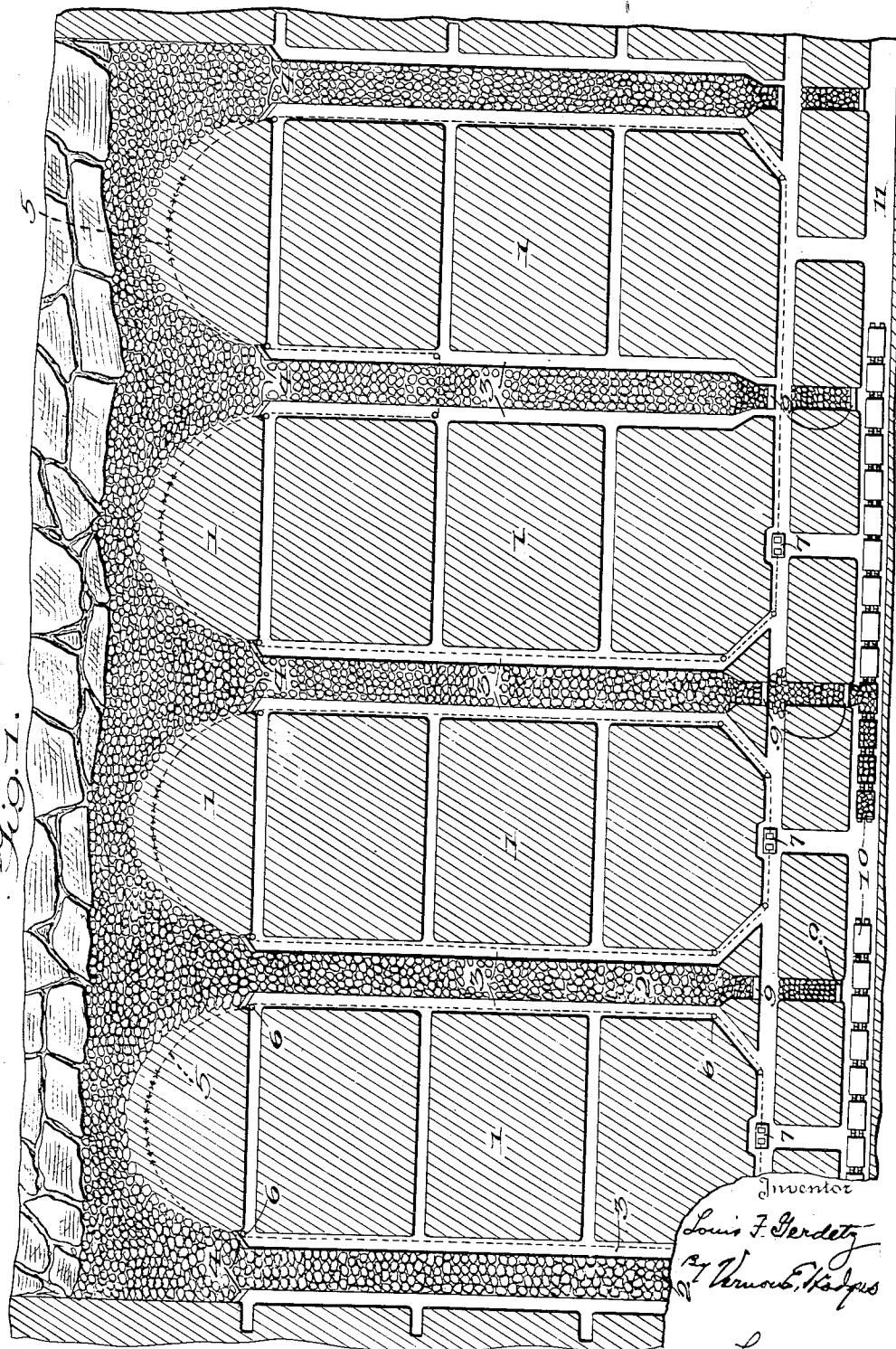
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METHOD OF MINING COAL

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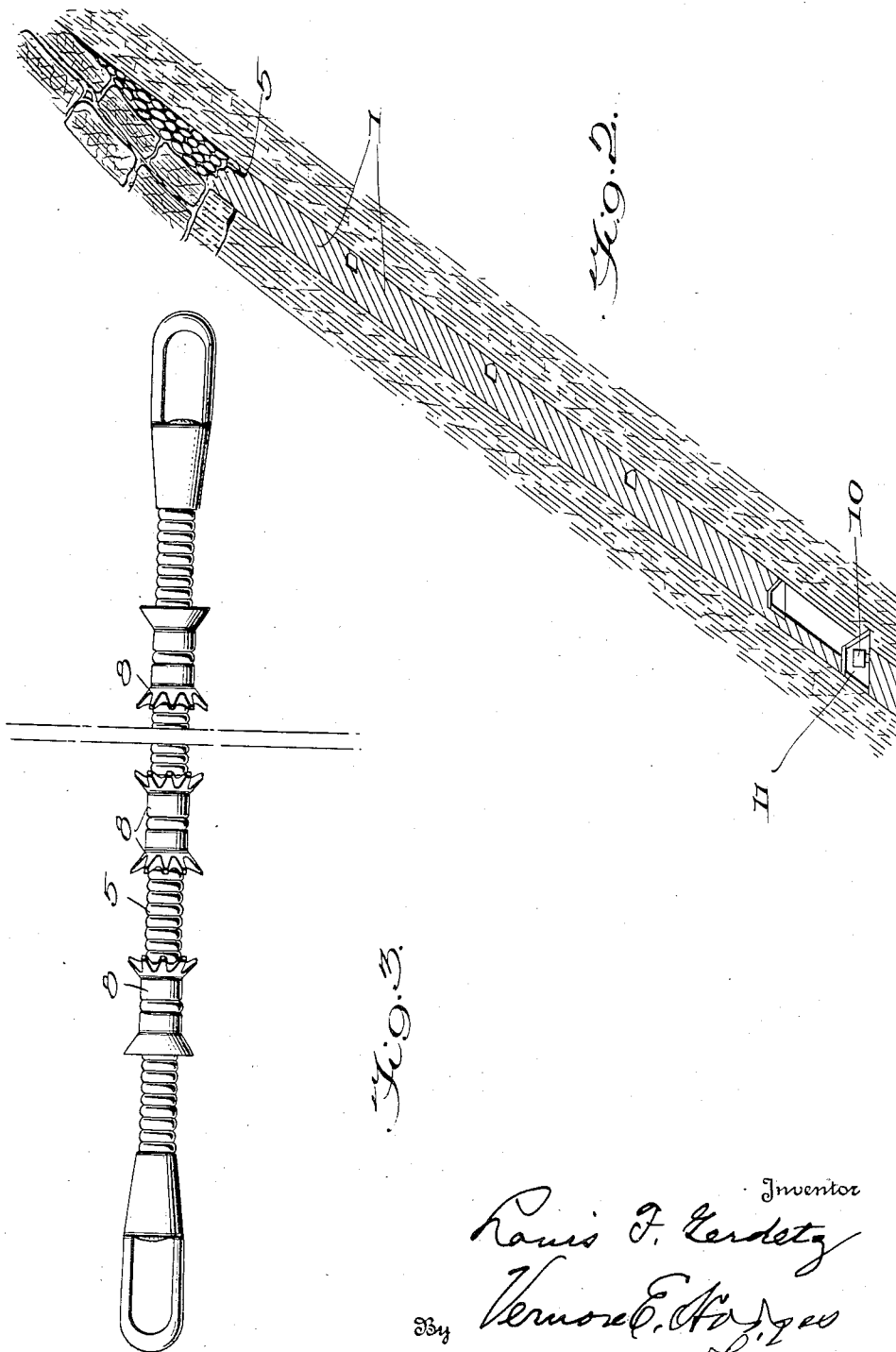


Fig. 3.

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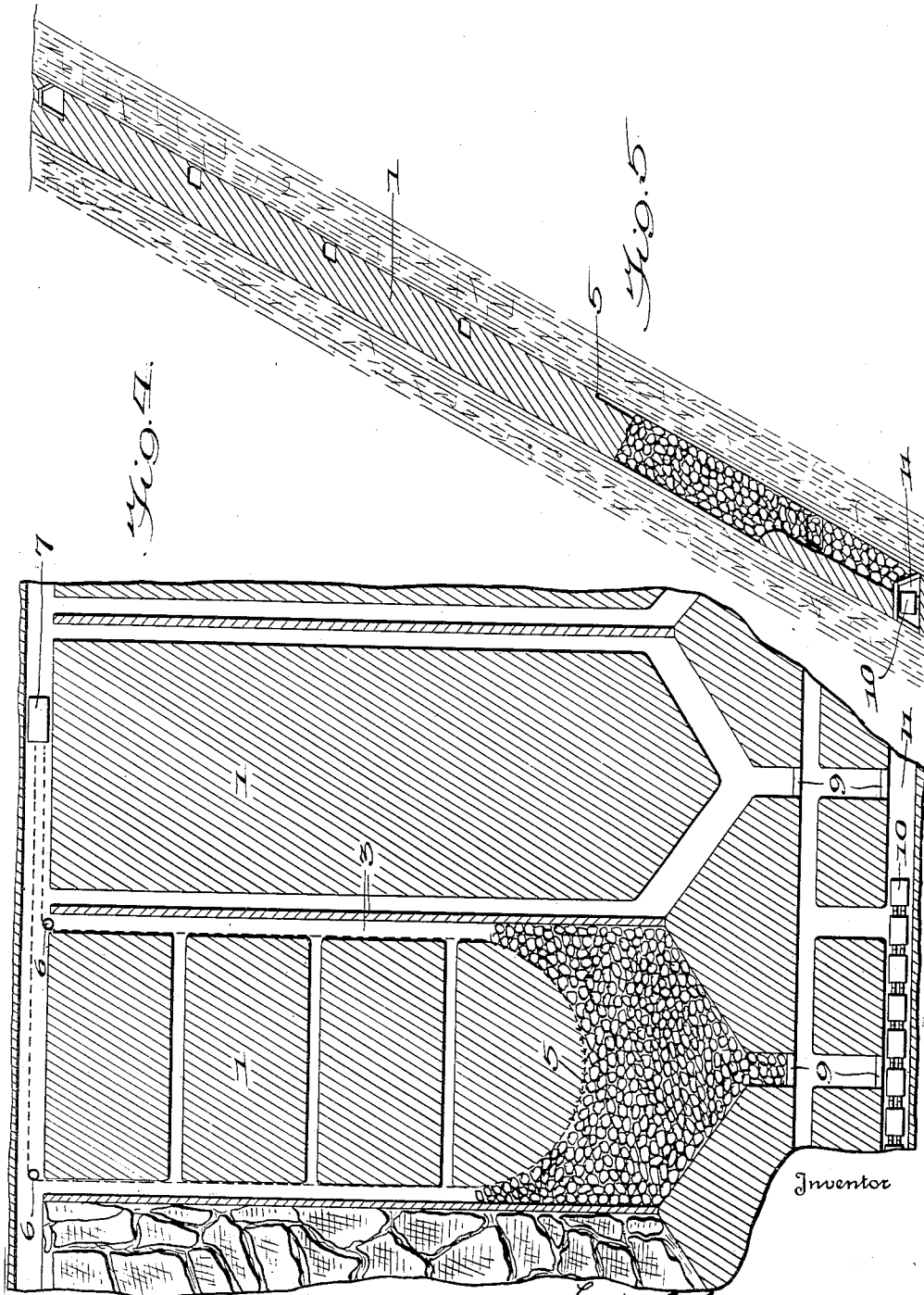
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METHOD OF MINING COAL

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16 Claims. (Cl. 262—1)

This invention relates to a method of mining coal, and more particularly to the mining of seams of coal which run in a generally oblique direction.

5 Coal-cutting machines have been employed heretofore for cutting coal where the seam extends in a generally horizontal plane, but where it is at a substantial acute angle to the horizontal such machines have not been very satisfactory, 10 for it is difficult to hold them in place during the cutting action. Also, in making cuts with a machine, it is only possible to cut a depth corresponding with the length of the cutter-bar, and then the cutting-machine must be backed off, and 15 the coal broken down and hauled out of the way of the cutter before another cut may be made.

In the pillar system of mining, a breast must be cut by hand labor, and heretofore these breasts have been spaced at only narrow intervals of 20 from thirty to sixty feet, leaving comparatively narrow pillars therebetween. Then by splitting the pillars with a pillar hole, additional coal is taken out, but this is all done by manual labor, and there is always a remaining pillar.

25 In my present invention, I utilize a system of undercutting the pillars which is applicable to pillars of at least twice the width which usually have been cut heretofore, thus reducing the number of breasts which must be cut by hand labor 30 between the pillars, and also eliminates pillar holes through the pillars. I may utilize a flexible cutter-cable with an attached cutter-chain which cable extends around the pillar and is reciprocated by a suitable reciprocating engine which 35 also takes up the slack to cause the chain to cut its way into the coal forming the entire pillar, which action may be continued until the entire pillar has been undercut and removed.

This system is particularly desirable and applicable 40 where the seam is arranged in a generally oblique or upright plane, and where it would be practically impossible to cut the coal by a cutting-machine. By undercutting the pillars in such an oblique seam, the coal breaks down 45 and flows down chutes automatically by gravity, and by arranging the cars or trucks at the ends of the chutes and suitably controlling the chutes, the coal is automatically loaded for transportation. Suitable manways are provided along the 50 sides of the pillars, so as to provide passageways for miners in changing the batteries or replacing a dulled cutter-chain.

This principle may be applied to cutting uphill, down-hill, or sidewise in a generally horizontal direction as may be found desirable; but

the direction of cut would not prevent the free downward flow of the cut and broken-down coal through chutes arranged according to the direction of cutting for directing the removal of the coal automatically from an inclined seam. 60

In the accompanying drawings:

Fig. 1 is a diagrammatic view of a portion of a mine illustrating an application of my improved method in making a downhill cut;

Fig. 2 is a vertical sectional view therethrough, 65 showing the oblique seam;

Fig. 3 is a side elevation of a cutter-chain with a part broken away, which may be used in making these cuts;

Fig. 4 is a diagrammatic view showing an uphill cut; 70

Fig. 5 is a vertical sectional view of the same.

As shown in Fig. 1 of the drawings, pillars of solid coal are designated 1 and are formed by cutting breasts 2 into the seam of the coal, which 75 breasts form gravity chutes between the pillars as applied to an oblique seam of coal to which this invention is particularly applicable. The pillars 1 are much wider than the pillars which have heretofore been left in cutting breasts be- 80 cause the breasts are much further apart. Manways 3 are provided along the sides of the pillars for the passage of miners therethrough, and the upper ends of these manways are closed by batteries 4 to prevent the coal which breaks down 85 from interfering with the manways. The manways 3 are intersected by the usual cross headings required for ventilation.

After the breasts are cut, forming pillars 1, my invention provides for undercutting the pillars, 90 and for this purpose I utilize an undercutting cable designated generally by the number 5, having a cutter-chain such as shown in Fig. 3, and which cable extends around the pillars and over suitable guide-pulleys 6, with its opposite ends 95 brought together and attached to a reciprocating engine 7, such as a well-known hoisting engine, provided with drums to receive the ends of the cable or ropes attached to the chain, and operated for reciprocating the cutting cable or chain, 100 while at the same time drawing in the slack to keep the cable taut. As the cutting cable is thus operated, it forms a kerf in the bottom of the pillar, undercutting the coal. After the kerf has penetrated the pillar to a substantial depth, the 105 weight of the coal or superimposed strata or roof will cause it to break down, or suitable means for breaking it down may be used, or a very slight charge of explosives will effectively accomplish the result. 110

The cutter-chain shown in Fig. 3 is of sufficient length to extend part way around the pillar, but does not extend to the reciprocating engine 7. Ropes, cables or chains may be attached to the opposite ends of the cutter-chain, and extend therefrom to the engine. In this form of cutter-chain, I have shown cutters 8 sleeved over a flexible cable, and freely rotatable thereon, so that the teeth cut their way into the coal as the chain is reciprocated. This chain is of the form set forth and claimed in my companion application on cutter-chains, and is set forth herein for purposes of illustration only.

After the coal is undercut and broken down, it flows by gravity down through the chutes 2 formed by the cutting of the breasts, and at the lower ends of these chutes loading batteries 9 may be provided to control the flow of coal from the chutes into cars or trucks 10, which operate in a passageway 11 extending transversely of the discharge ends of the chutes.

In this way the complete cut is made without the necessity of having any men at the point of cutting, only a man to operate the engine, but even this may be set to operate automatically, so that the cutting operation will be carried out absolutely without manual labor beyond the necessary inspection and changing of cables. When it is necessary or desirable to remove the cutter-chain for sharpening, after it has become dulled, the cutting operation may be stopped and the entire cutter-chain easily detached, and a sharp cutter-chain substituted therefor, in order that the dulled chain may be taken out and the cutters sharpened. After the cutter-chain has cut into the coal a substantial distance, the protective batteries of the manway may be reset from the inside of the manway without danger.

Usually the undercutting cable will extend around only one pillar, as shown at the left side of Fig. 1, but if the reciprocating engine is sufficiently strong it may extend around and undercut two pillars at a time, as shown at the right side of Fig. 1.

Fig. 2 shows an oblique seam of coal to which this invention is particularly applicable, for it enables the coal to be undercut, broken down, and to flow down by gravity through the chutes formed by the breasts and thus eliminate a great amount of handling and substantially reducing the danger of cutting coal in a seam of this character. The cost of removing coal from such a seam is reduced a substantial extent by the use of this invention, in addition to the greater degree of safety involved.

While I have shown in Figs. 1 and 2 the method of cutting downhill to remove the coal from an oblique or inclined seam, yet, if desired, the cutting operation may be sideways, or upwards. The latter is illustrated in Figs. 4 and 5, which is practically the reverse of the downhill cutting action above described, except that a single chute extends downward from the middle of each pillar for discharge of the coal. Only narrow manways are provided between the pillars divided by a partition, as shown, having spans beside the pillars for the passage of miners to inspect or replace a cutter-chain. The broken down coal fills the passages and chutes around the pillars beneath the point of cutting, tending to prevent the falling through of the superimposed strata and roof. The discharge from the chutes is controlled to provide for this. After the entire pillar has been undercut, or so much of it as desired, and broken down, the discharge closures

for the chutes may be released, allowing the coal to flow downward by gravity. The weight of the superimposed strata and roof on the broken down coal aids in forcing the coal out.

Additional kerfs may be cut in the pillars at the top and/or intermediate points, or vertical cuts may be made after boring holes at the top and bottom of the seam to receive the cable. In fact, a square cut may be made in the seam by such vertical cuts at the sides and top and bottom cuts, thus cutting out a complete block of coal.

I claim:

1. A method of mining coal or the like disposed in an inclined seam, comprising forming a pillar in the seam, forming an inclined chute or chutes in the coal, and undercutting the pillar to break down the coal and direct it downward by gravity through the chute.

2. A method of mining coal or the like disposed in an inclined seam, comprising cutting inclined openings through the coal with a pillar therebetween, undercutting the pillar from one side to the other thereof to break down the coal in the pillar and direct the coal downwardly by gravity through the inclined breasts.

3. A method of mining coal or the like disposed in an inclined seam, comprising cutting inclined breasts through the coal with a pillar therebetween, undercutting the pillar from one side to the other thereof, breaking down the coal in the pillar, directing the coal downwardly by gravity through the inclined breasts, regulating the discharge of the coal from the breasts into conveying receptacles to be hauled off thereby.

4. A method of mining coal or the like from an inclined seam, comprising cutting breasts in the seam approximately throughout the height thereof, and in the inclined plane of the seam, forming a pillar extending approximately throughout the height of the seam, undercutting the pillar from one end thereof to the other to break down the coal and automatically discharge it by gravity through chutes formed by the breasts.

5. A method of mining coal or the like normally disposed in an inclined seam, comprising cutting an inclined chute in the seam and approximately throughout the height of the seam, forming a pillar adjacent said chute, operating an undercutting cable around the pillar to undercut the pillar from one side thereof to the opposite side in an inclined plane extending lengthwise of the pillar to break down the coal, discharging the coal by gravity through the chute.

6. A method of mining coal or the like normally disposed in an inclined seam, comprising cutting inclined breasts in the seam and approximately throughout the height of the seam, forming a pillar therebetween, directing an undercutting cable around the pillar to operate the cable to undercut the pillar from one side thereof to the opposite side to break down the coal and direct the coal through the breasts by gravity, controlling the discharge from the breasts, and discharging the coal from the breasts by gravity into a conveying receptacle.

7. In machinery for undercutting a pillar of coal or the like arranged with openings on opposite sides thereof, the combination with manways arranged along the sides of the pillar, batteries closing the ends of the manways, of an undercutting cable extending entirely around the pillar near the base thereof, and carrying cutting tools on the portion thereof adjacent a side

of the pillar to cut a kerf in the pillar, means connected with said cable at the opposite side of the pillar from the cutting tools for causing reciprocating motion of the cable and drawing the same into the kerf formed by the tools, and pulleys arranged in the manways to receive and guide the undercutting cable therethrough.

8. A method of mining coal from an inclined seam cut to form a pillar therein, comprising cutting a natural gravity chute in the coal extending downwardly from a side of the pillar, and beginning at said side gradually undercutting at least the major portion of the pillar to break down the coal and flow the same by gravity downward through said natural chute.

9. A method of mining coal from an inclined seam cut to form a pillar, comprising cutting a natural gravity chute in the seam of coal with its mouth located at the bottom of a side of the pillar, and beginning at said side gradually undercutting at least the major portion of the pillar to break down the coal and flow the same automatically by gravity downward through said natural chute.

10. A method of mining coal from an inclined seam cut to form a pillar therein, comprising cutting a natural gravity chute in the seam of coal with its mouth approximately at a side of the pillar, moving a cutter transversely across said side of the pillar near the base thereof to undercut the pillar with a kerf and break down the coal at the mouth of the chute into and through which it flows by gravity.

11. A method of mining coal from an inclined seam cut to form a pillar therein, comprising cutting a natural gravity chute in the seam of coal with its mouth approximately at a side of the pillar, moving a cutter transversely across said side of the pillar near the base thereof to undercut the pillar with a kerf and break down the coal at the mouth of the chute into and through which it flows by gravity, continuing said undercutting action until approximately the entire pillar has been undercut and broken down, and controlling the discharge from the chute, keeping the chute and spaces around the pillar approximately full of coal below the point of cutting until the cutting action has progressed to a desired point.

12. A method of mining coal from an inclined

seam cut to form a pillar therein, comprising cutting a natural gravity chute in the seam of coal extending downwardly from the lower end of the pillar, and beginning at said lower end of the pillar gradually projecting a transverse undercutting kerf at the base of the pillar projected from said lower end gradually toward the upper end approximately at the oblique angle of the seam to break down the coal and cause it to flow downward by gravity through the chute away from the point of cutting.

13. A method of mining coal from an inclined seam cut to form a pillar therein, comprising undercutting the pillar from side to side, and from an end projecting said undercut inwardly of the pillar at an oblique angle approximately corresponding with the oblique angle of the inclined seam to allow the coal to break down and flow by gravity from the seam.

14. A method of mining coal from a pillar of an inclined seam comprising undercutting the pillar at an oblique angle, permitting the coal to break down throughout the undercut portion to flow by gravity from the undercut area, and permitting the overlying strata to settle down on the broken down coal to force the same from its broken down position.

15. A method of mining coal from an inclined seam comprising boring oblique holes at relatively low points in the seam to receive a cutting cable for undercutting the coal, passing a cutting cable into said bored holes and at least partially around a portion of the coal therebetween, and operating said cables to make an oblique undercut in the coal permitting the undercut coal to break down by gravity.

16. A method of mining coal from an inclined seam comprising boring oblique, approximately parallel and relatively small holes at relatively low points in the seam with a cross opening joining the corresponding ends of the bored holes, passing a cutting cable through the bored holes and the cross opening, at least partially around a portion of the coal between the oblique holes, operating said cables to make an oblique undercut in the coal between said bored holes, and permitting the overlying strata to settle down breaking down the undercut coal and to force the broken down coal from its position.

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